

FORENSIC LOG ANALYSIS REPORT

HIGH-FIDELITY EVIDENCE INVESTIGATION & THREAT INTELLIGENCE

CASE METADATA & EVIDENCE RECORD

Incident Case ID:	FLX-20260609-190212
Date of Report:	2026-06-09 19:02:12
Lead Analyst:	Gohar Ali
Target Organization:	BS Digital Forensics [amp] Cyber Security
Evidence File Source:	demo_apache_access.log
Evidence Hash (SHA-256):	a3f8c2d91e4b7f0612a85c3d9e2f1b4a7c0d3e6f8a1b4c7d0e3f6a9b2c5d8e1f

1. Executive Summary & Overview

Automated security auditing of the evidence source `demo_apache_access.log` has been completed. The forensic ingest engine processed a total of **875** log entries, originating from **10** distinct network endpoints. A total of **5** malicious signatures or behavioral anomalies were flagged. Based on the severity of the threat telemetry, the overall security posture is classified as **CRITICAL RISK**. Immediate containment and security controls tuning are recommended. The following sections provide detailed technical logs, MITRE ATT&CK; technique mappings, and structured remediation roadmaps aligned with NIST SP 800-61 incident response phases.

TOTAL REQUESTS AUDITED	SECURITY THREATS FOUND
875	5
UNIQUE ATTACKING IPS	POSTURE RISK INDEX
10	CRITICAL RISK

2. Evidence Integrity & Chain of Custody

A rigorous chain of custody and cryptographic hash record has been maintained to ensure the admissibility, authenticity, and non-repudiation of the digital evidence under analysis.

#	Timestamp	Action Taken	Action Details / Audit Log	Authorized Actor
1	2024-03-15 10:00:00	Evidence Collected	demo_apache_access.log	Gohar Ali
2	2024-03-15 10:00:01	Hash Computed	SHA-256: a3f8c2d91e4b...	System
3	2024-03-15 10:00:02	Log Ingestion Complete	875 lines processed	System
4	2024-03-15 10:00:03	Threat Intel Correlated	5 alerts triggered	System

3. Indication of Compromise (IoC) & Threat Analysis

The following table details the specific threat signatures, Web Application Firewall (WAF) alerts, and heuristic detections triggered during log inspection. Detections are ordered by severity (critical/high prioritized).

#	Threat Category	Severity	Source IP	Count	Evidence Details Summary
1	Brute Force	CRITICAL	185.220.101.34	84	84 auth failures from 185.220.101.34
2	HTTP Flood / DDoS	HIGH	91.108.4.180	65	65 req/min — possible DDoS
3	Directory Traversal	HIGH	45.33.32.156	15	Path traversal: ../../[slash]etc/[passwd]
4	Vulnerability Scanner	MEDIUM	91.108.4.180	30	Scanner: Nikto — 30 requests
5	Weak Session IDs	MEDIUM	10.0.0.22	42	Predictable session token sequence

4. MITRE ATT&CK; Framework Mapping

To align with security standards and industry best practices, detected threat categories are mapped below to standard MITRE ATT&CK; Techniques, IDs, and Tactics.

Threat Category	MITRE Technique	Technique ID	Associated Tactic
Brute Force	Brute Force	T1110	Credential Access
HTTP Flood / DDoS	Network Denial of Service	T1498	Impact
Directory Traversal	File and Directory Discovery	T1083	Discovery
Vulnerability Scanner	Active Scanning	T1595	Reconnaissance
Weak Session IDs	Modify Authentication Process	T1556	Credential Access / Defense Evasion

5. NIST-Aligned Actionable Roadmap

Remediation actions are structured below in accordance with the **NIST SP 800-61 r2** Computer Security Incident Handling Guide, prioritizing immediate containment followed by eradication and hardening.

Response Phase	Target Action Items & Implementation Details
PHASE 1 Containment (0-24 Hours)	• Block Malicious IPs: Update firewall rules (iptables/UFW) to drop incoming packets from identified attack sources: {} • Terminate Active Sessions: Invalidate current active sessions or tokens associated with the suspicious IP ranges. • Isolate Vulnerable Endpoints: Limit access to targeted URL paths (/wp-login.php, /admin, config files) to trusted internal IPs.
PHASE 2 Eradication (24-72 Hours)	• Patch Vulnerable Web Apps: Update CMS core files, plugins, and server scripts to eliminate code-level flaws (SQL Injection, Path Traversal). • Deploy Rate-Limiting: Configure nginx/Apache limit_req modules or fail2ban rules to rate-limit authentication routes. • File Integrity Scans: Run deep file-integrity checks (SHA-256) across public HTML paths to detect unauthorized code modifications.
PHASE 3 Strategic Hardening (Post Incident)	• Deploy Web Application Firewall (WAF): Fully enable ModSecurity and tune the OWASP Core Rule Set to active blocking mode. • Centralized Logging: Move log auditing to a SIEM system (e.g. Wazuh, ELK Stack) to enable real-time detection and alerting. • Regular Audits: Establish a weekly automated forensic review schedule using ForensicLogX with cryptographic verification.